Theorem 7(A+1=1): This theorem states that when a variable A is ORed with 1, the output is always equal to1. One input to an OR gate is always 1 and the other input A can be either 1 or 0. Now, 1 on one of the inputs to the OR gate produces 1 on the output regardless of the value of the variable on the other input.

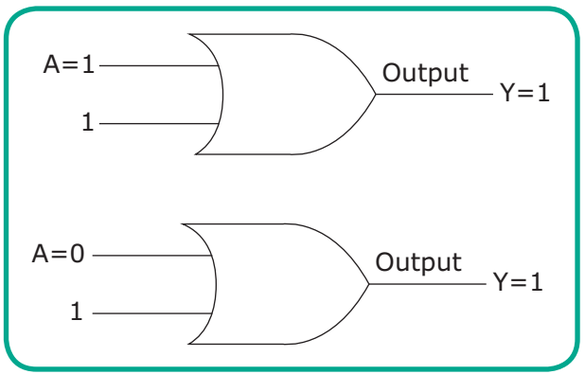


Figure 8.9 Logic gate representation of Theorem 7

Theorem 8(A.0=0): This theorem states that variable A ANDed with 0 always produces 0. Recall that any time one input to an AND gate is 0, the output is 0 regardless of the variable A on the other input.

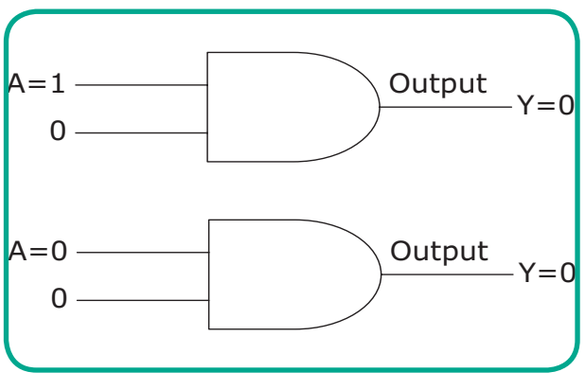


Figure 8.10 Logic gate representation of Theorem 8

Theorem 9($\overline{\overline{A}}$=A): This theorem states that if a variable A is complemented twice, the result is the variable itself. Starting with A and inverting (complementing) it once gives $\overline{A}$. Inverting it once more, gives $\overline{\overline{A}}$ the original value.

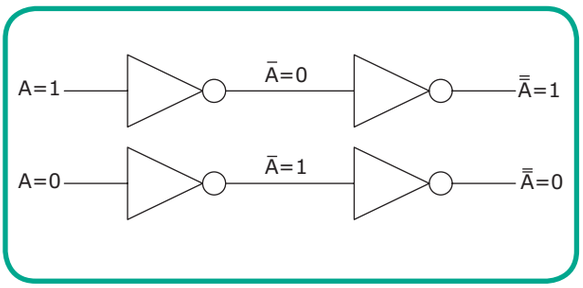


Figure 8.11 Logic gate representation of Theorem 9

8.9 ADVANTAGES AND DISADVANTAGES OF DIGITAL ELECTRONICS

8.9.1 Advantage

1. Very simple logic the lead to identify the faults very easily
2. Immune to noise
3. Flexibility of programming
4. Design and testing is very simple compared to analog electronics
5. Achieve very high speed switching

Disadvantage

1. High energy consumption than analog electronic circuits
2. Higher cost of design
3. Portability is difficult
4. Real world signals need conversion
5. Less accurate than the analog electronics

"One of the very important characteristics of a student is to question. Let the students ask questions."

-A.P.J. Abdul Kalam

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